

YANNY OR LORA

CREATING AN EMERGENCY RADIO SYSTEM USING LORA TECHNOLOGY





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WHO WE ARE



- Eric Hamber Secondary School — Vancouver, Canada
- **Asa Liu (co-captain)** — LoRa development, hardware, frame
- **Matthew Song (co-captain)** — landing gear
- **Zoey Cheang** — LoRa protocol, software
- **Naiming Zheng** — software
- **Dora Yuan** — outreach, landing gear
- **Bogdan Shkromiuk** — hardware
- Teacher advisor: Mr. Mark Lam



THE PROBLEM



- Many areas, like BC (where we're from) and Alberta (here!) can have long wildfire seasons
- Wildfires, floods, and natural disasters destroy cell towers and ground infrastructure
- Hikers and people in remote areas **have no reliable communication**
- Existing solutions (Starlink) cost \$600+ and require setup — not viable in emergencies
- Ground-level radios have limited range
- **Our question: how to develop an emergency communication system that *anyone* can easily deploy, without spending \$\$\$ or relying on existing infrastructure?**



OUR SOLUTION :)



- We built a CanSat that passes emergency messages between two devices (home base + emergency device) that can't reach each other directly
- Uses LoRa (Long Range) technology
- A spider web network system of our LoRa CanSats would provide extensive communication across several kilometers
- Low cost, no infrastructure required, deployable by anyone in minutes



OUR MISSION

The Concept

- A person in a remote location sends an SOS message via a small LoRa device
- CanSat receives the message while airborne
- CanSat relays the message to a home base receiver kilometres away
- Home base would be unreachable from the sender without the aerial relay
- Message contains: SOS signal, GPS coordinates, heartbeat sensor data





OUR MISSION

- Our CanSat also fulfills the **standard primary mission**
- Collect temperature, air pressure, and humidity every second during descent
- Sensor: BME280 (temperature, pressure, humidity)
- Data transmitted every second via LoRa radio to ground station





HARDWARE



- **LilyGO TTGO T-Beam** — single board containing:
 - **ESP32 microcontroller** (the brain)
 - **SX1276 LoRa radio** (communication)
 - **NEO-6M GPS** (location tracking)
 - Built-in LiPo battery management
- **BME280** — temperature, pressure, humidity
- **915MHz LoRa antenna** — communication with ground stations
- **LiPo battery** — mounted at bottom to lower centre of mass
- 3x T-Beams total: one in can, one emergency device, one home base



SYSTEM ARCHITECTURE



3 device system

- **CanSat (airborne):** receives from emergency device → stores → rebroadcasts to home base
- **Emergency device (ground):** transmits SOS + GPS + heartbeat
- **Home base (ground):** receives relayed message, logs and displays data



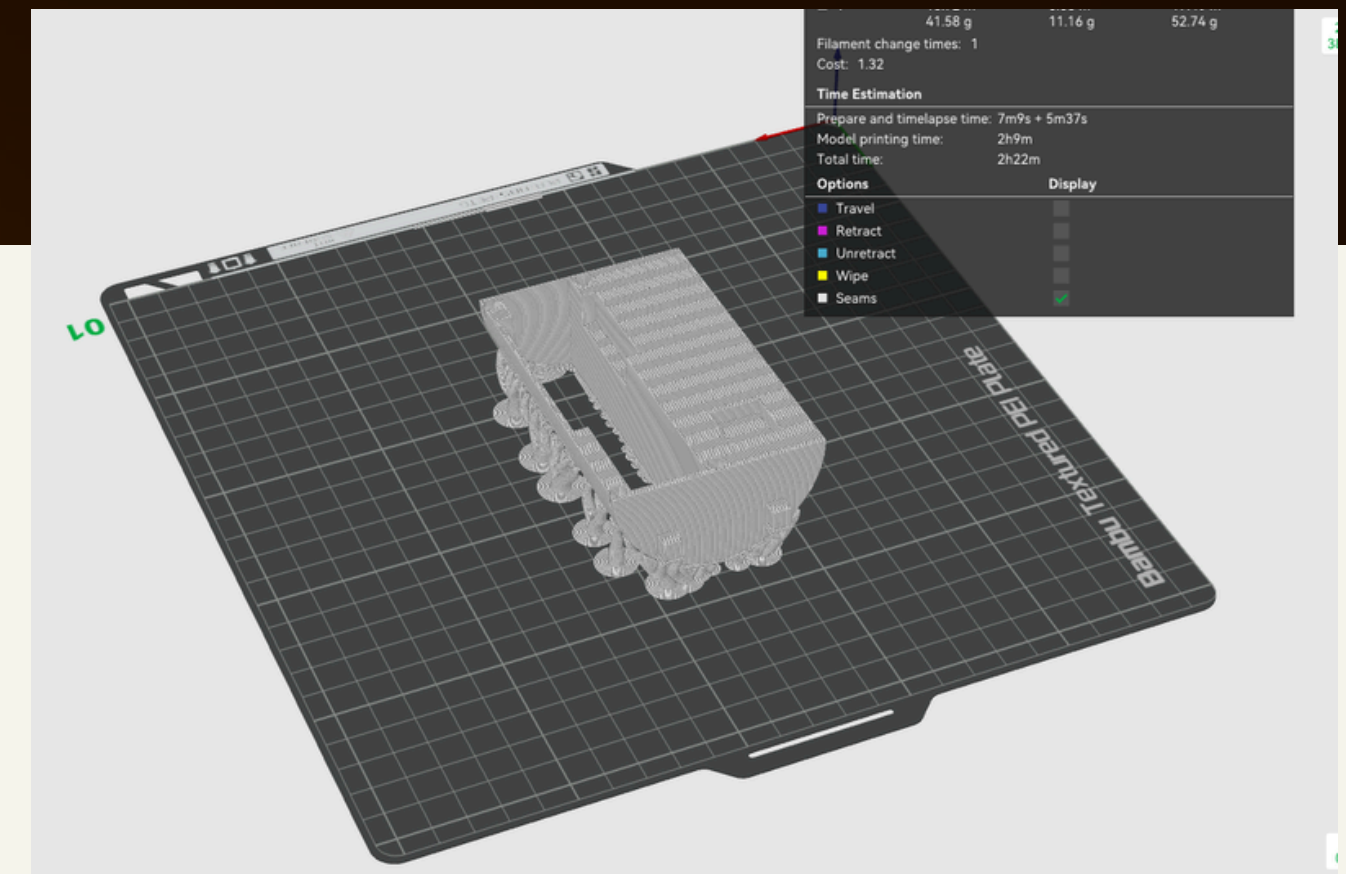
FRAME TESTING

Goal: Create a frame able to withstand 20gs of force, fitting all hardware and remaining under the size limit

Setup: We decided to use PLA with a gyroid infill for distribution of force while remaining light and flexible. After testing with weights and the parachute we decided on 80% infill for top and bottom lids and 20% infill for the body

Results: Our first 5 versions of testing concluded in many failures but excelled in stability and impact force reduction.

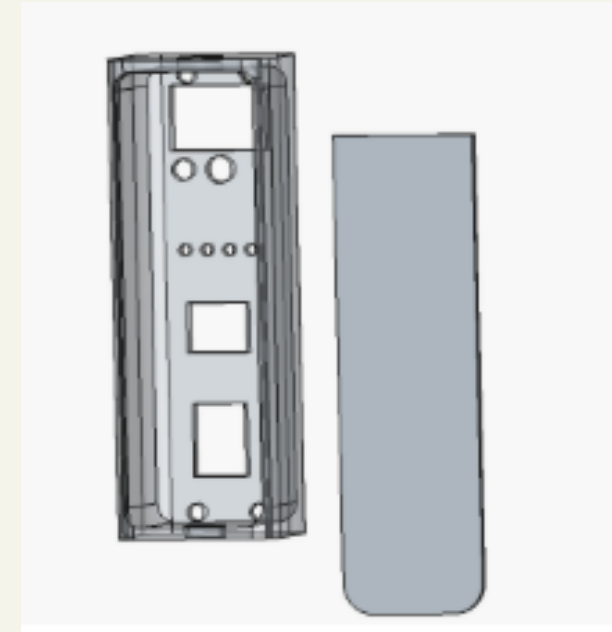
Issues: Most of our failures was due to lack of space. The main issue was fitting the antenna and the wires within the given dimentions. We started with hexagons and pentagons but realized we weren't benefitting from the full range so we ended back with circles.



SOS RADIO FRAME

The SOS Radio Frame was built with a few things in mind.

1. Has to be able to fit; Antenna, BME280 sensor, oled screen, heartbeat sensor, buzzer, and LoRa32
2. Must be compact and accessible
3. Must be strong enough to not break under hiking environment.



Version 1

- Was too small to fit parts
- Had incorrect measurements
- Printed with 20% Infill PLA

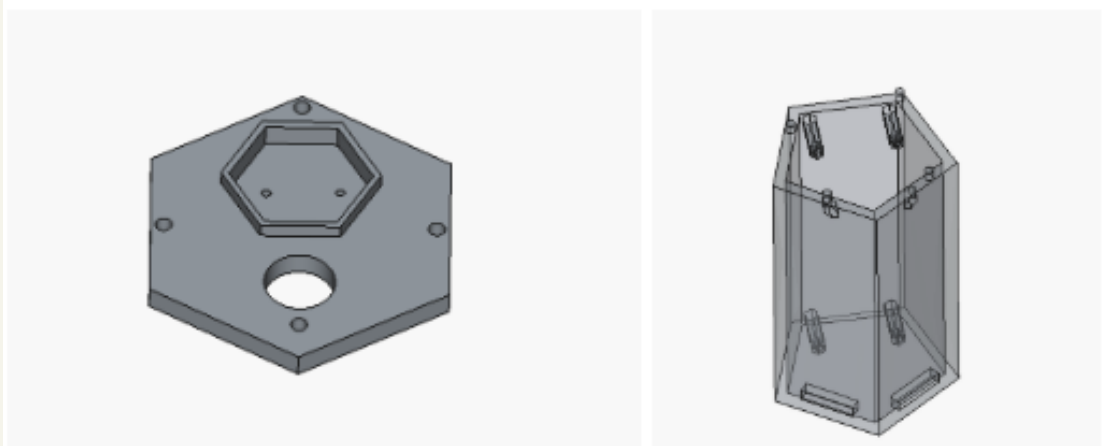


Version 2

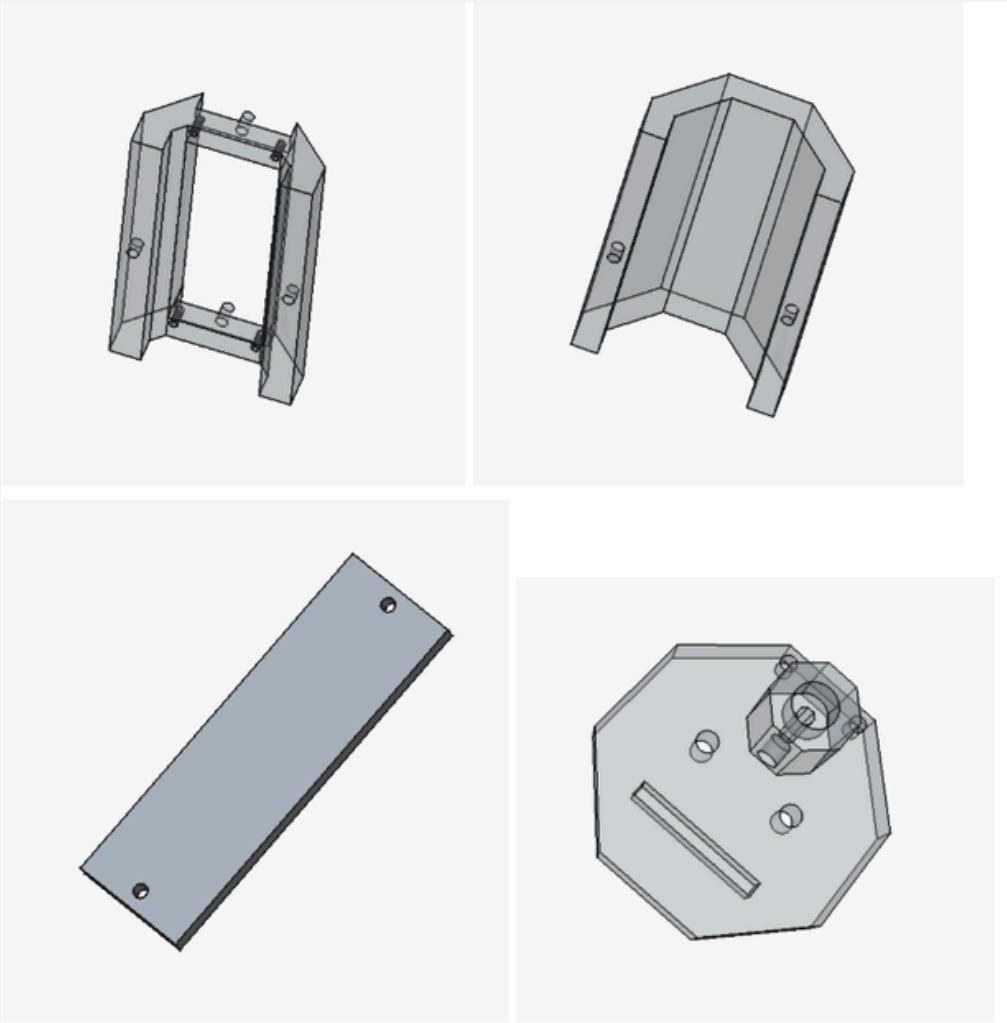
- Fit everything inside
- Was a little oversized
- Printed with nylon (stronger)

PROTOTYPES

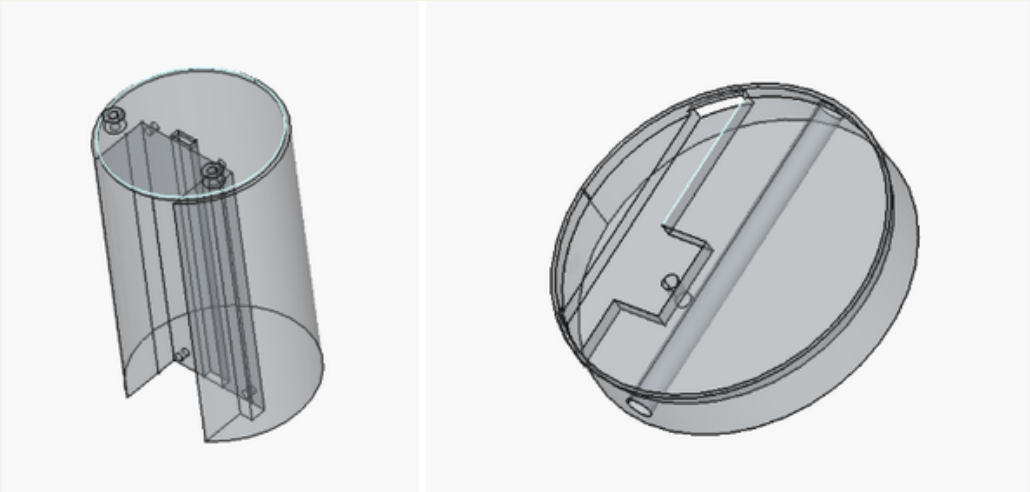
Version 3



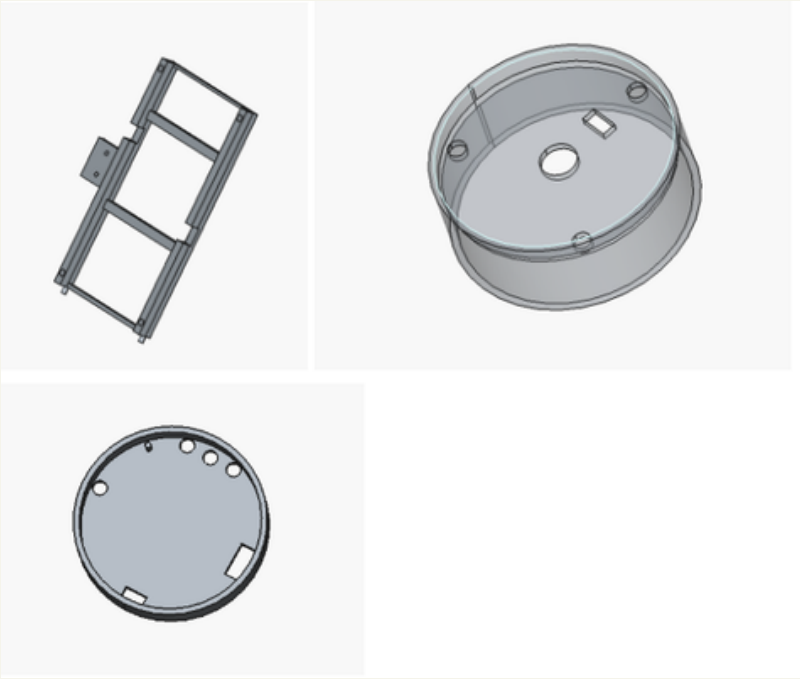
Version 4



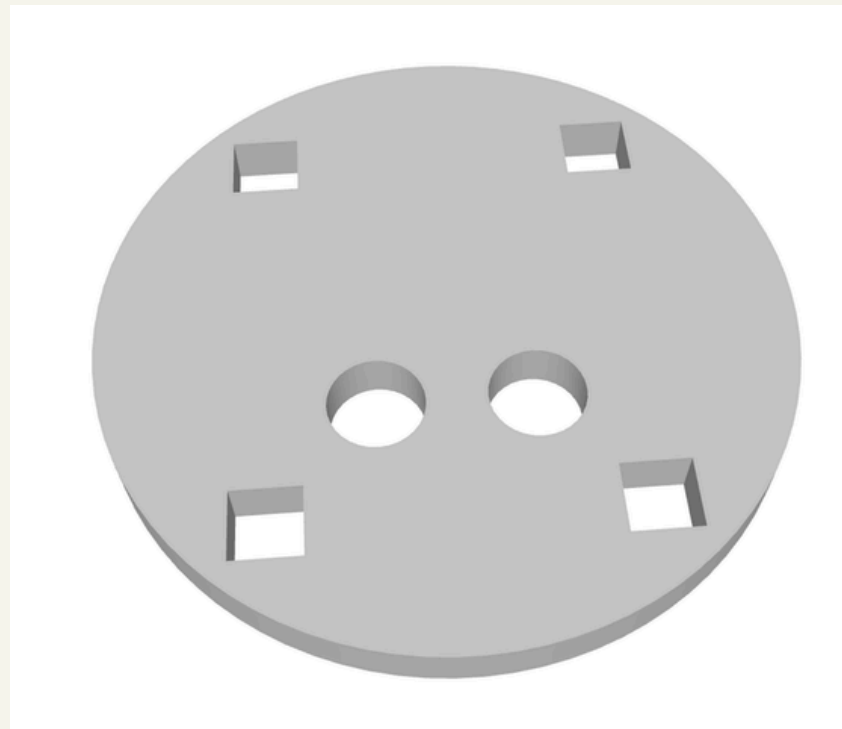
Original Version



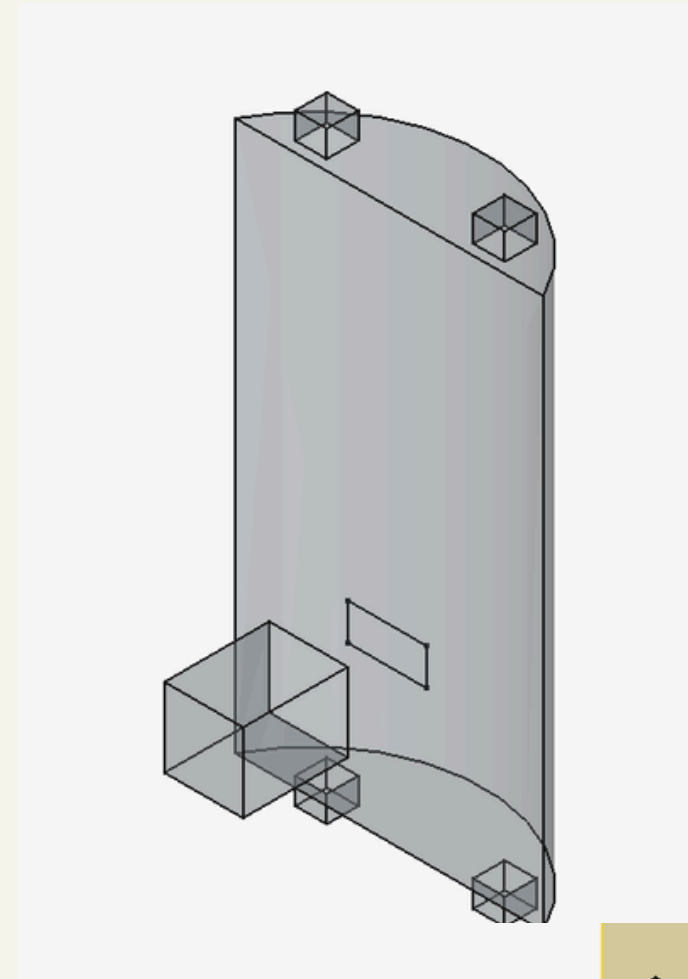
Version 2



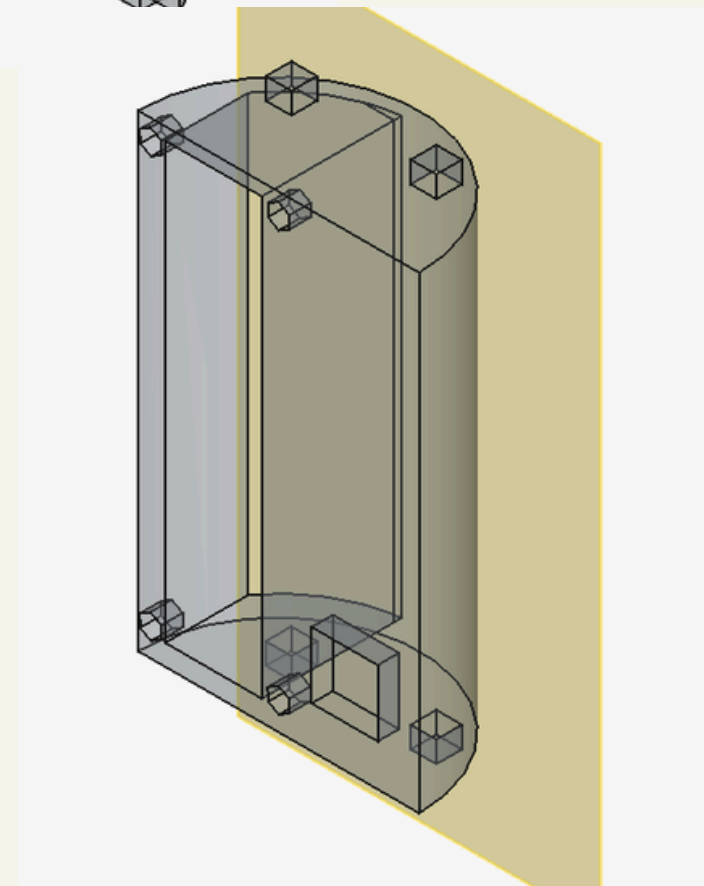
FRAME



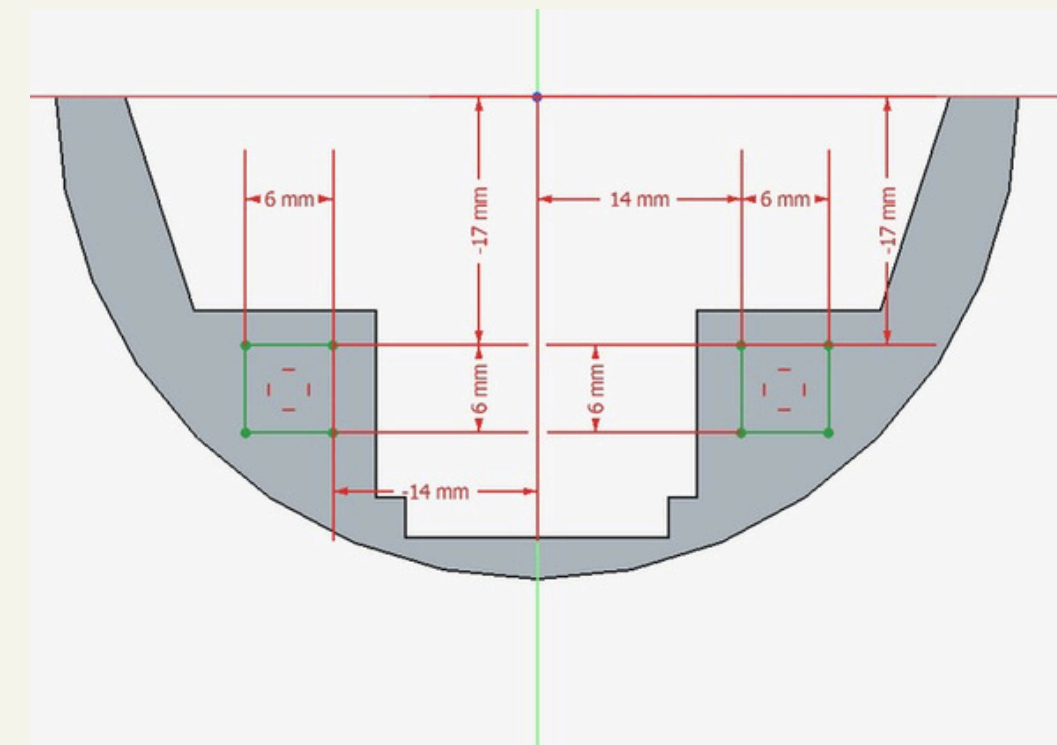
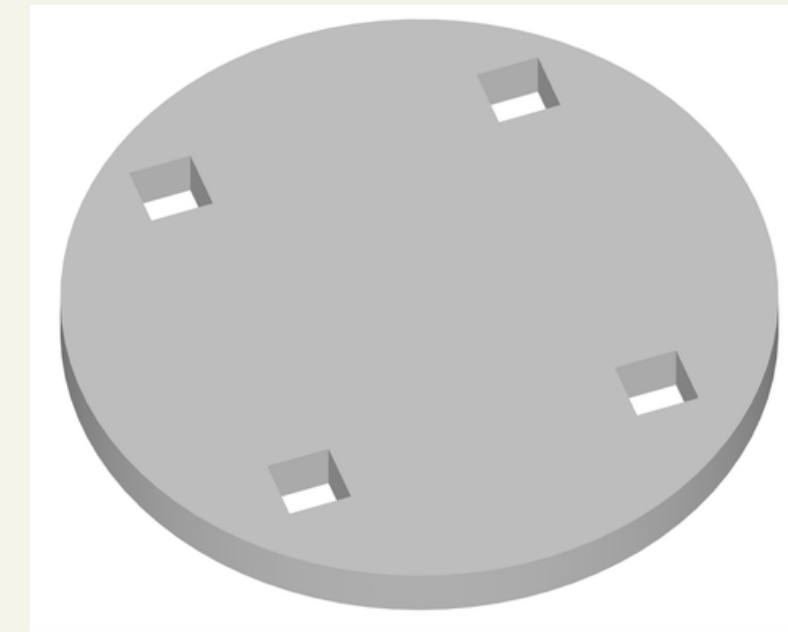
Top lid



Body



Bottom

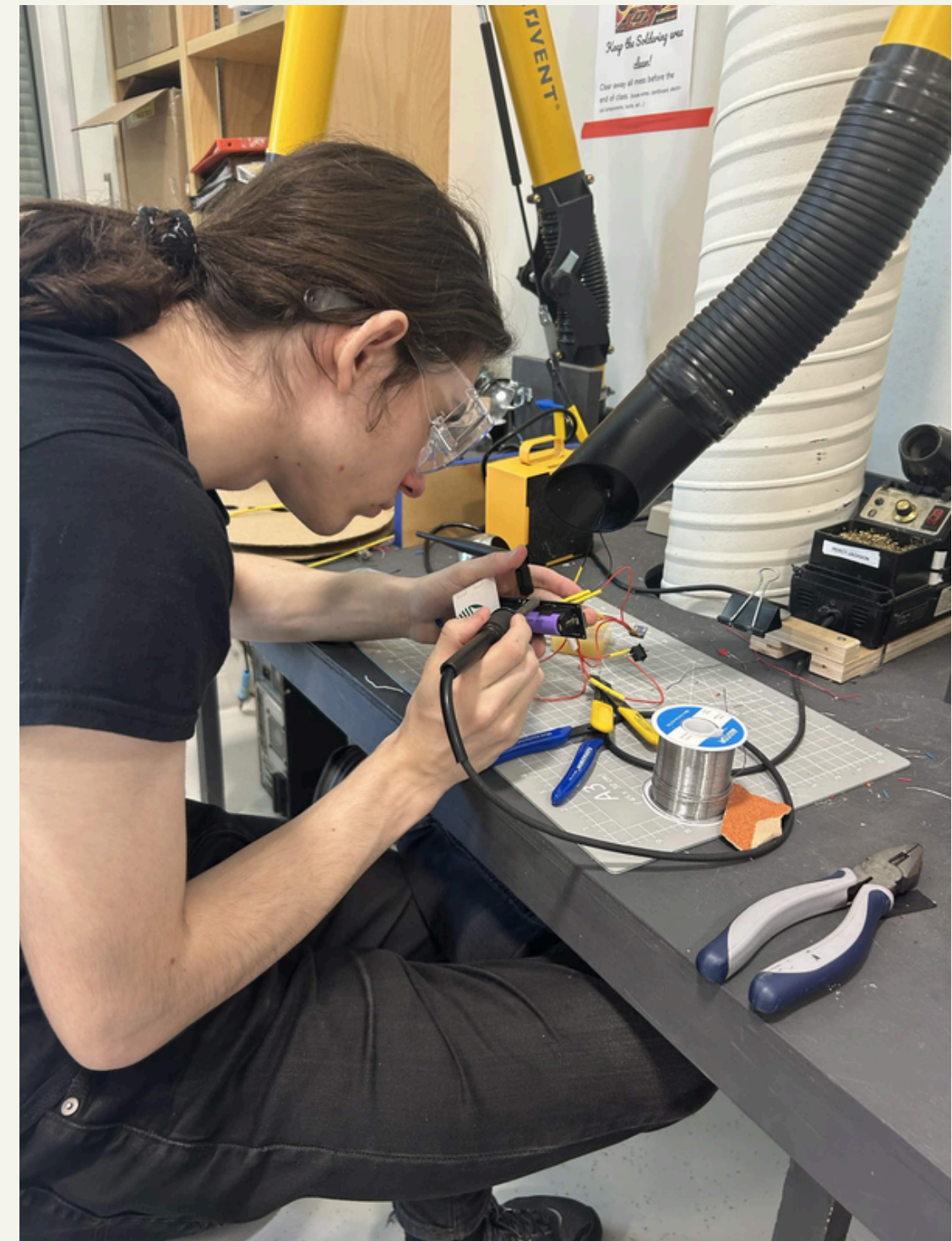




ELECTRICAL DESIGN

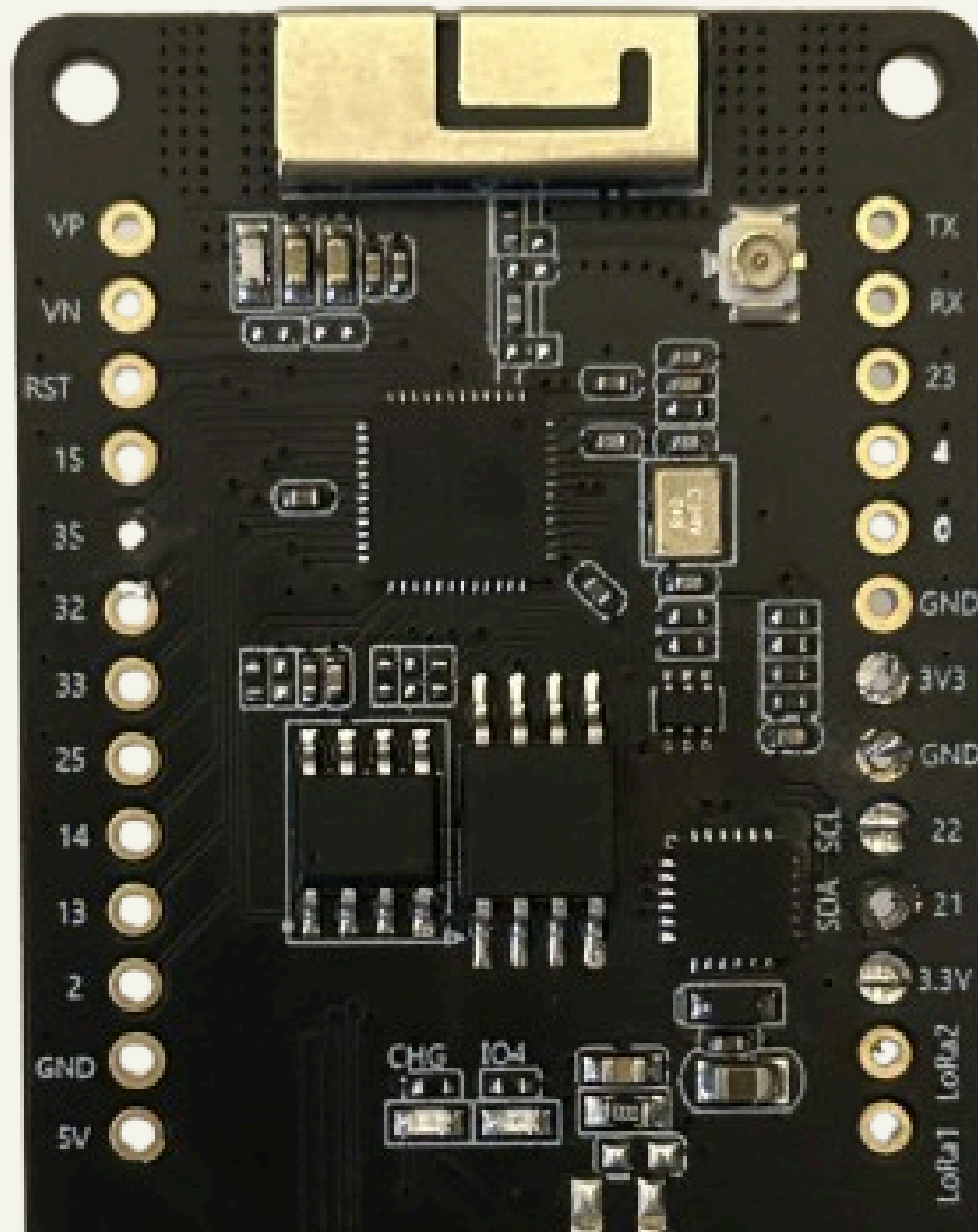
Three devices:

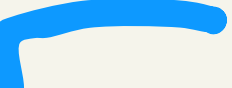
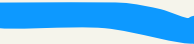
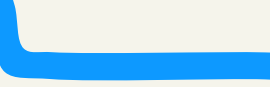
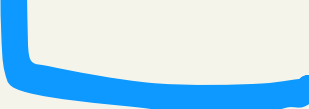
- SOS radio device
- CanSat
- Ground Station





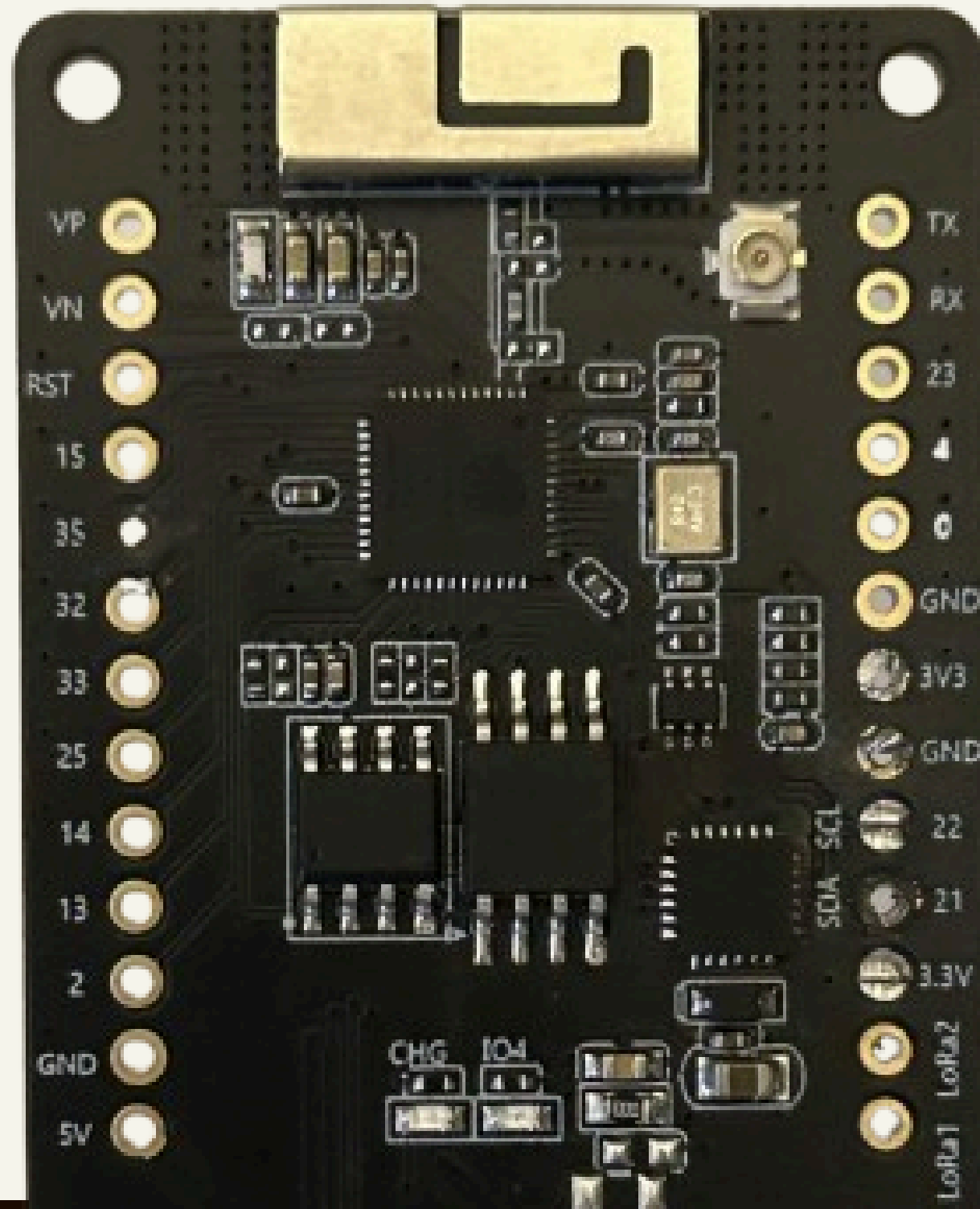
SOS RADIO DEVICE WIRING



- 3.3V  OLED screen (SDA=21, SCL=22)
-  BME 280 (SDA=21, SCL=22, addr 0x76/0x77)
-  MAX30102 (SDA=21, SCL=22)
- GND  OLED
-  BME280
-  MAX30102
-  Buzzer (+ leg to 5V)
-  SOS Button (other leg → GPIO 2, INPUT_PULLUP)



CANSAT WIRING



3.3V

BME280 (SDA=21, SCL=22, addr 0x76/0x77)

+ Leg LED

Power Switch (Other leg → GPIO 32)

GND

(-) Leg LED

BME280



SOS DEVICE FEATURE

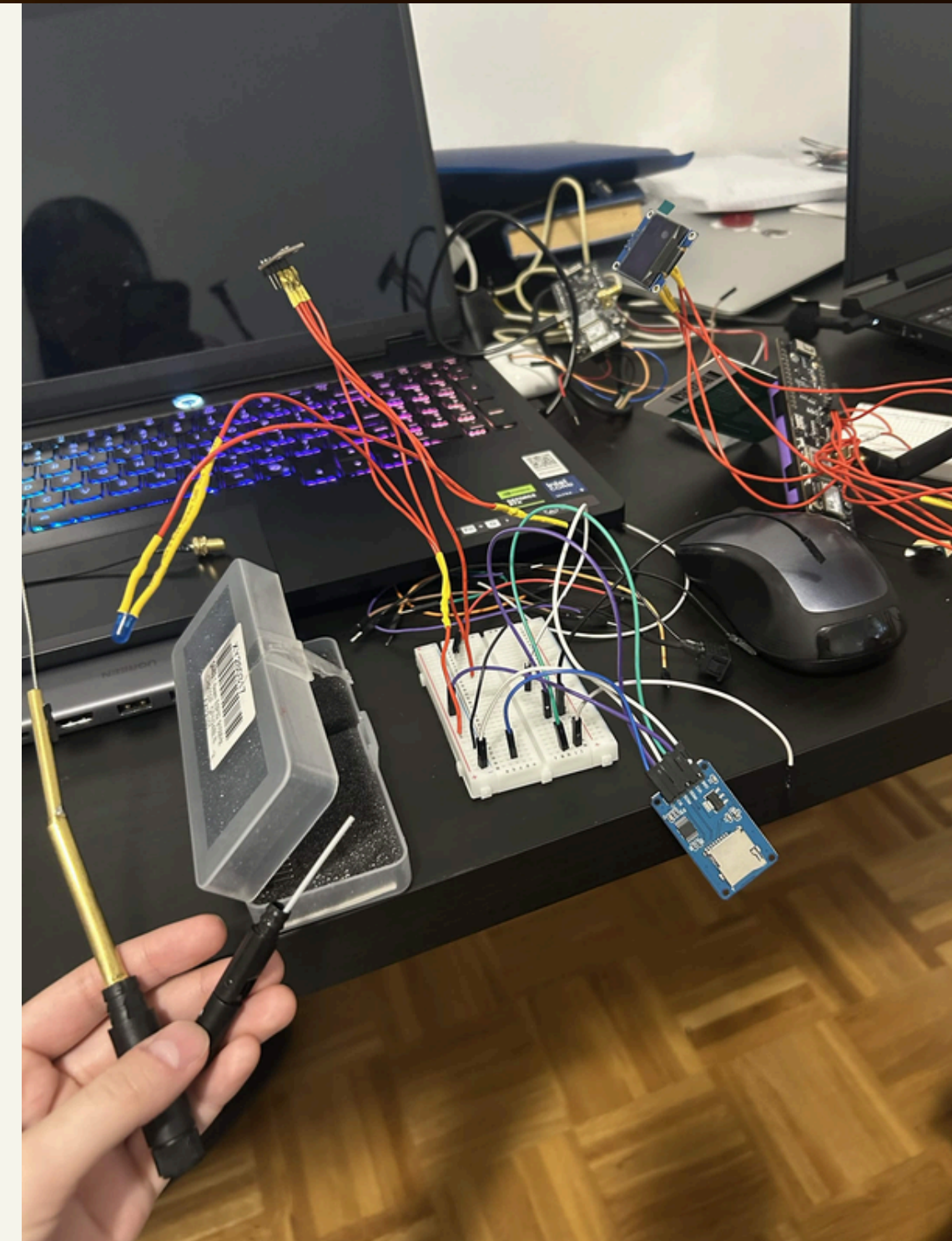
- OLED screen
- Heartbeat sensor (MAX 30102)
- Buzzer
- BME280
- GPS (NEO-6M)
- SOS Button
- Power switch





SOS DEVICE FEATURE

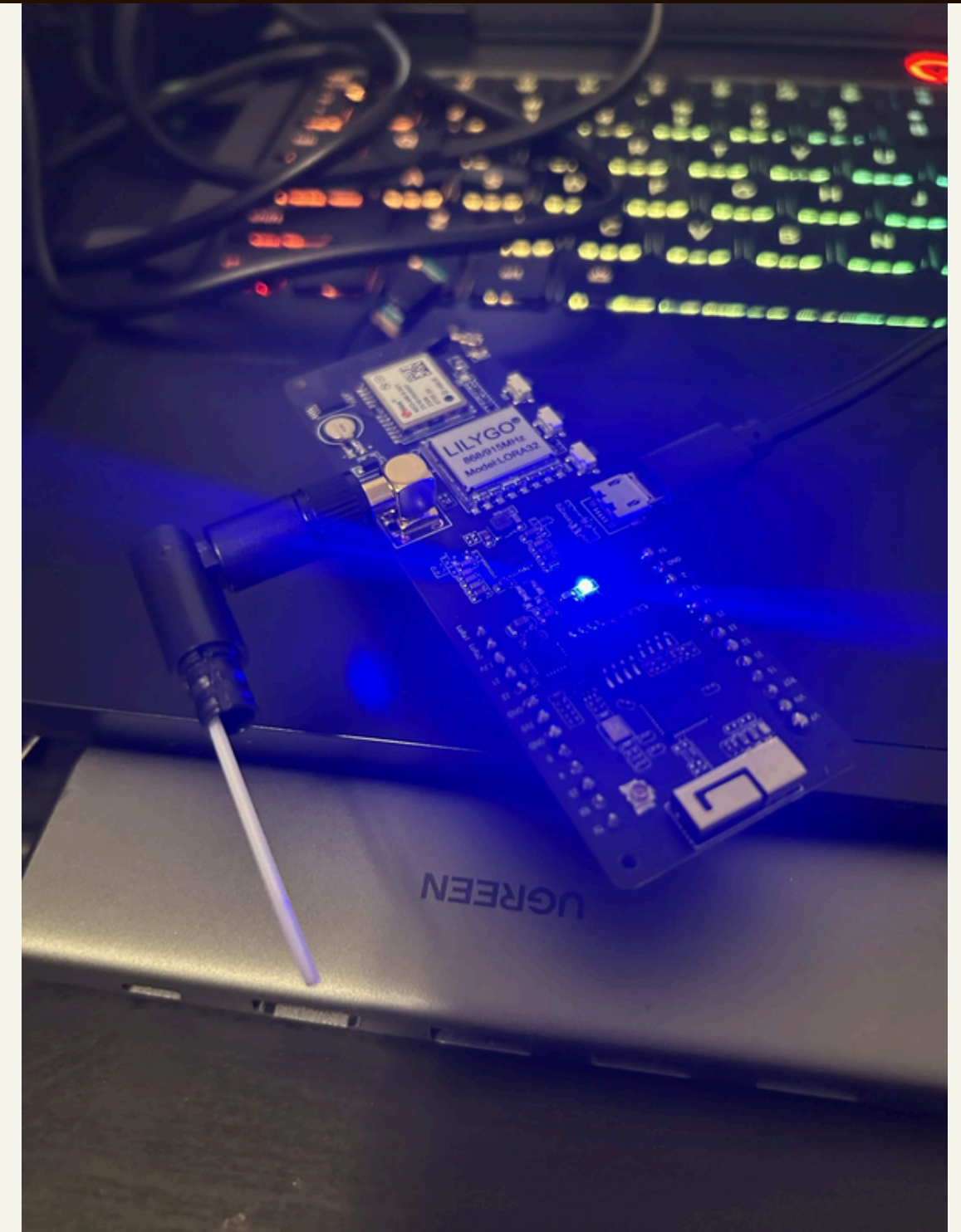
- OLED screen displays BME280 data (Altitude, Humidity, Pressure, Temperature), location, and heartbeat
- When finger is detected on the heartbeat sensor, OLED screen displays your heartbeat (BPM)
- Double click SOS button to send SOS message to CanSat
- After receiving ACK from CanSat, OLED screen shows the message send successfully





COMMUNICATION LOGIC

- LoRa device cannot receive and send data at the same time
- CanSat alternating between receiving (TX) mode and sending mode (RX)
- Sending data every seconds
- RX mode in around 0 – 850ms
- TX mode in around 850 – 910ms
- SOS radio device spam SOS message for 5 seconds
- CanSat send ACK to SOS after successfully receiving message
- TX mode (ACK if available) 910 – 925ms
- GPS, LED
- 80% of time in RX (ensure SOS data receiving successfully)

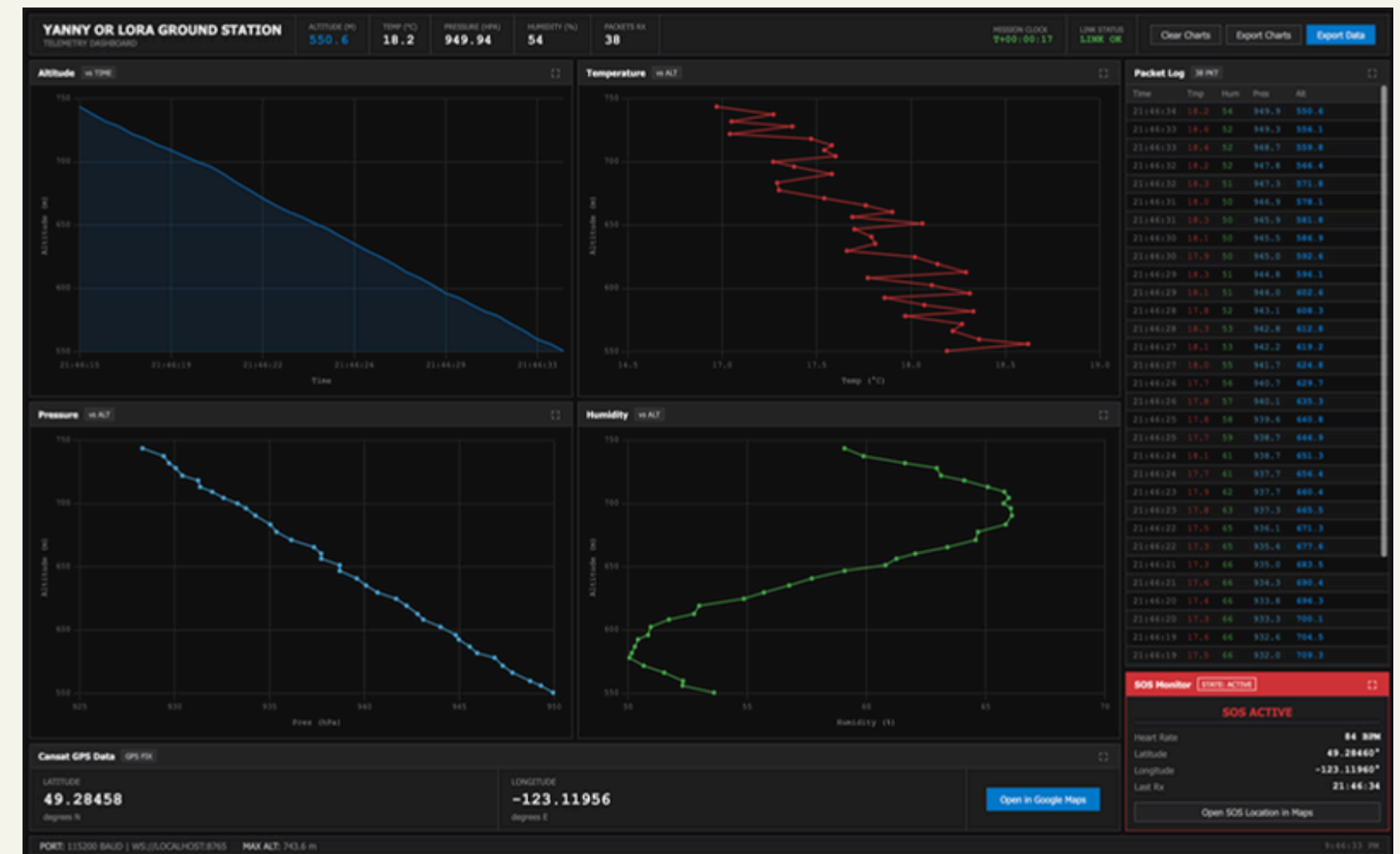




SOFTWARE

Features

- Temperature/pressure/humidity graphs + data
- Heartbeat + GPS coordinates + SOS signal
- We couldn't obtain a license for a static map on time, but we hope to do that for future prototypes





RECOVERY SYSTEM

- Circular parachute is our recovery system
- Has to withstand 30 Newtons of force, then descend at a rate of 5-12 m/s
- 8 strings of polypropylene twine (durable and hard to tangle) connect the parachute to the CanSat
- Parachute made of nylon fabric





PARACHUTE CALCULATIONS

$$F_D = \frac{1}{2} \rho v^2 C_d A$$

- Circular parachute
 - Circular parachutes have a drag coefficient of around 1.75
 - Plug in values to drag force equation. Estimated terminal velocity: approximately 5 m/s, can change due to wind



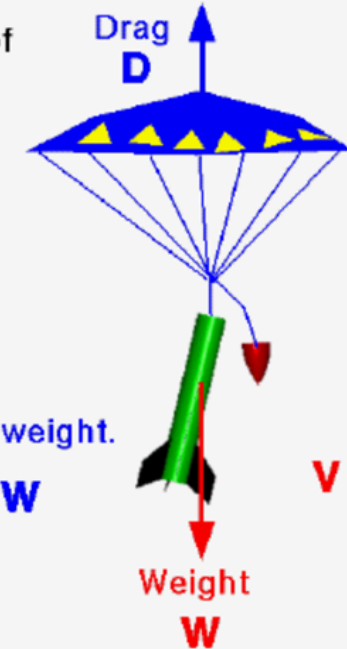
Velocity During Recovery

Glenn Research Center

Cd = drag coefficient of parachute
Cd = 1.75 (typical)

Drag Equation:
$$D = C_d \frac{\rho V^2}{2} A$$

During recovery, drag=weight.
$$D = C_d \frac{\rho V^2}{2} A = W$$



Weight
W

r = air density
r = 1.229 kg / cu m
A = parachute area
V = velocity

Solve for Velocity:
$$V = \text{sqrt} \left(\frac{2 W}{C_d r A} \right)$$



OUTREACH PROGRAM

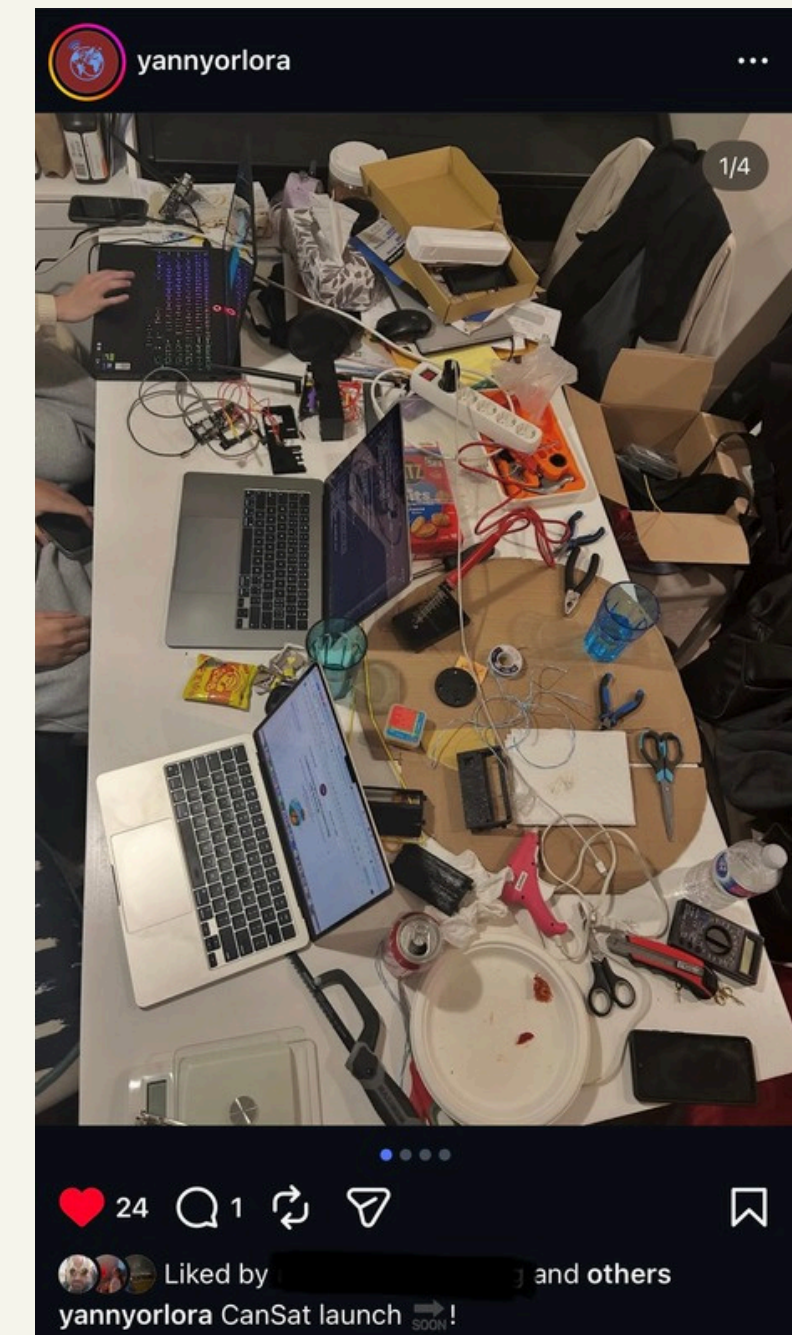
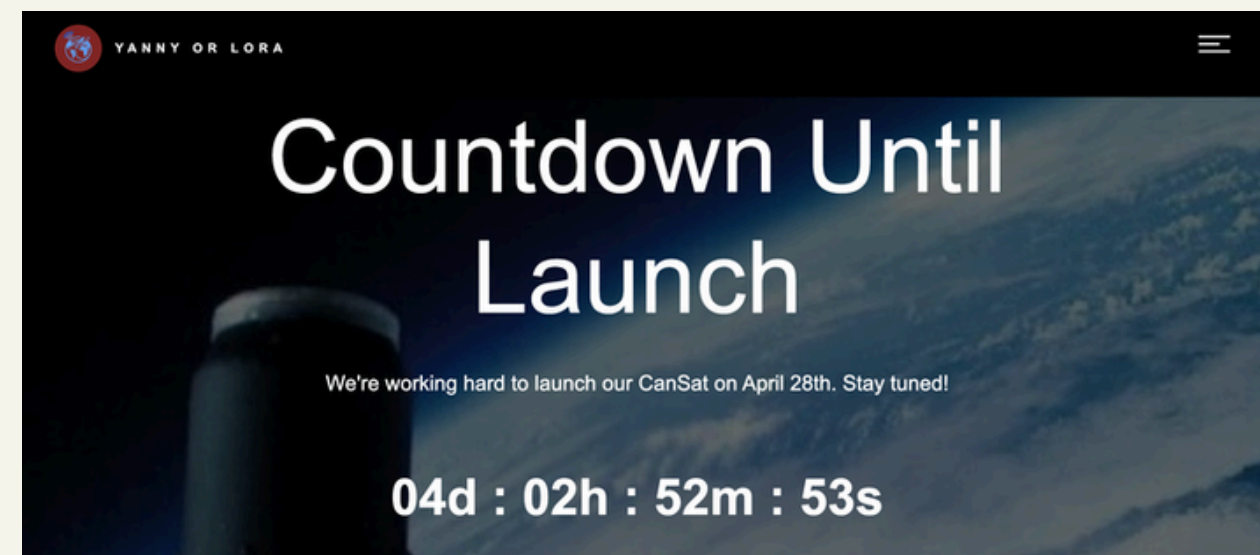
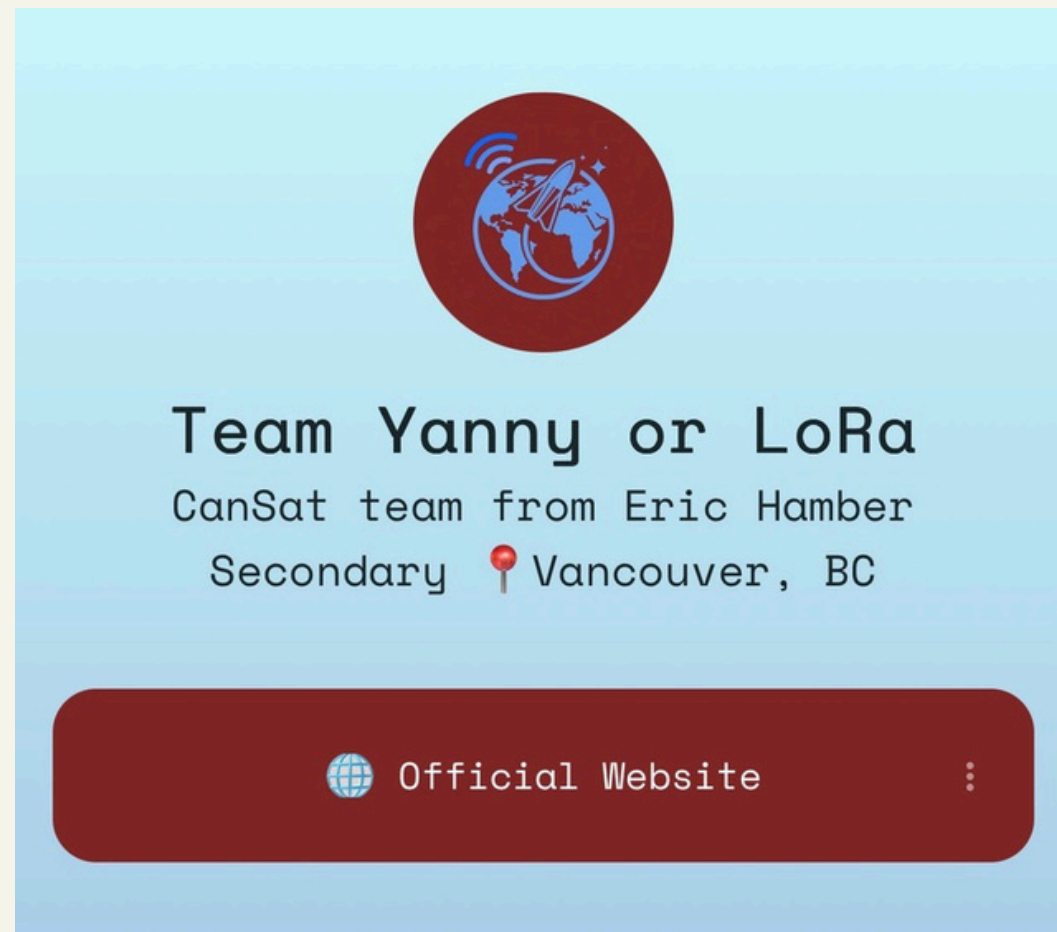


- Both social media and in person presentations

Platform	Link	Description
Website	https://yannyorlora.tech	CanSat information + mission homepage
Instagram	https://www.instagram.com/yannyorlora	Work-in-progress updates (on our story), informative posts, record of in-person outreach activities
LinkTree	https://linktr.ee/yannyorlora	Central hub for social media information

SOCIAL MEDIA

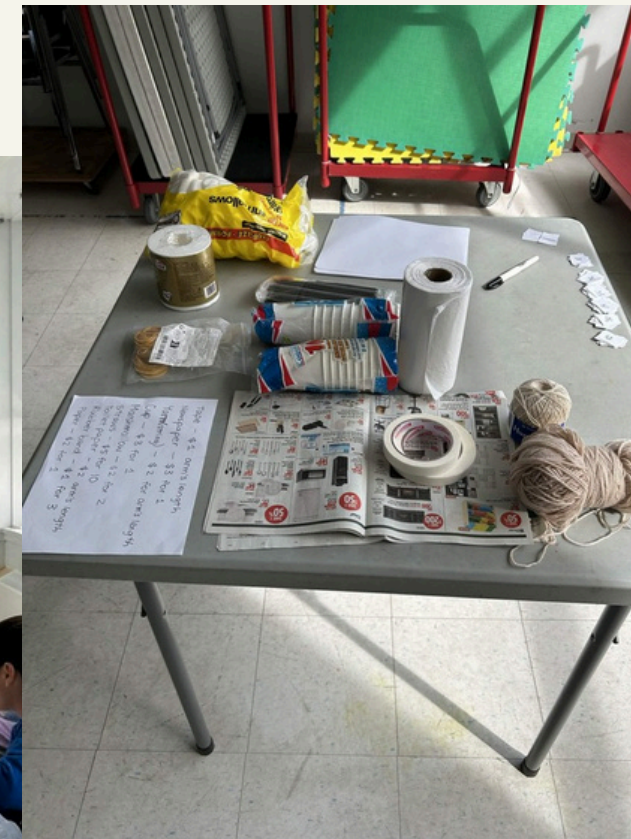
- What our social media looks like :)





OUTREACH PROGRAM

- 1st in person presentation: local community centre (**Kerrisdale Community Centre**)
- Class of kids aged 9-12 in a STEM program
- Presentation + egg drop





OUTREACH PROGRAM



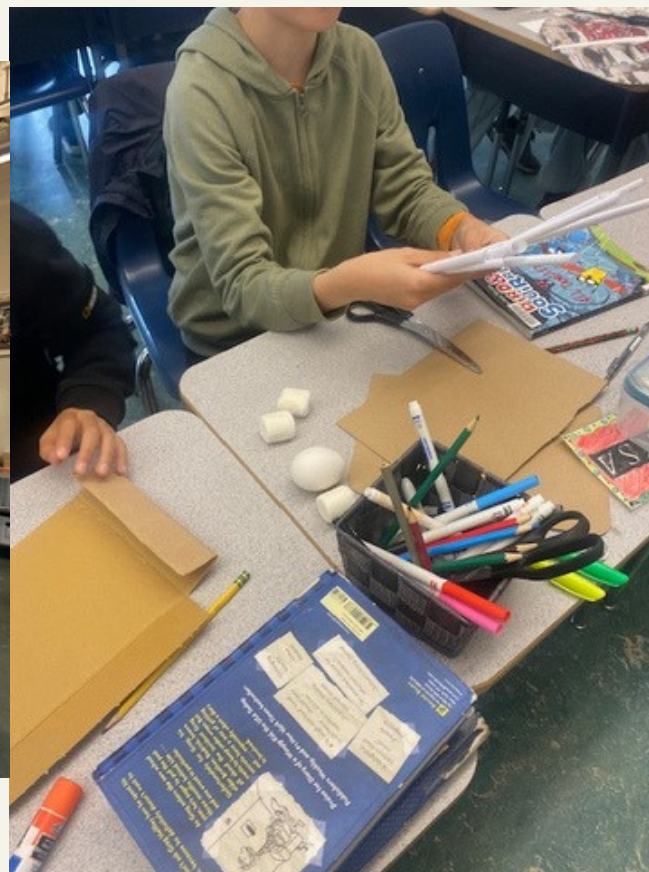
- 2nd in person presentation: **high school**
- STEM-focused school club called Experiment Club
- Audience very fitting because participants were already interested in engineering
- Circuit + breadboard activity
- Several of our participants expressed interest in the CanSat competition!!





OUTREACH PROGRAM

- 3rd in person presentation: **elementary school**
- Grade 4 class in Quilchena Elementary
- Presentation + egg drop





BUDGET

Component	Quantity	Cost per Item (CAD)	Total Cost (CAD)
LILYGO T-Beam V1.2 ESP32 LoRa Board	3	\$40	\$120
BMP280 Temperature & Pressure Sensor	1	\$0	\$0
905 MHz LoRa Antenna	3	\$15	\$45
LiPo Batteries	2	\$10	\$20
Battery Charger/Protection Module	1	\$8	\$8
Toggle Switch + LED Indicator	1	\$5	\$5
Wires / Connectors	-	-	\$5
PLA Filament (3D Printed Frame)	-	-	\$10
Parachute Materials	-	-	\$10
Small Hardware (screws, mounts)	-	-	\$10
USB Cables/Adapters	-	-	\$5
Antenna Mount/Stand	-	-	\$10
Total Cost: \$240 CAD			





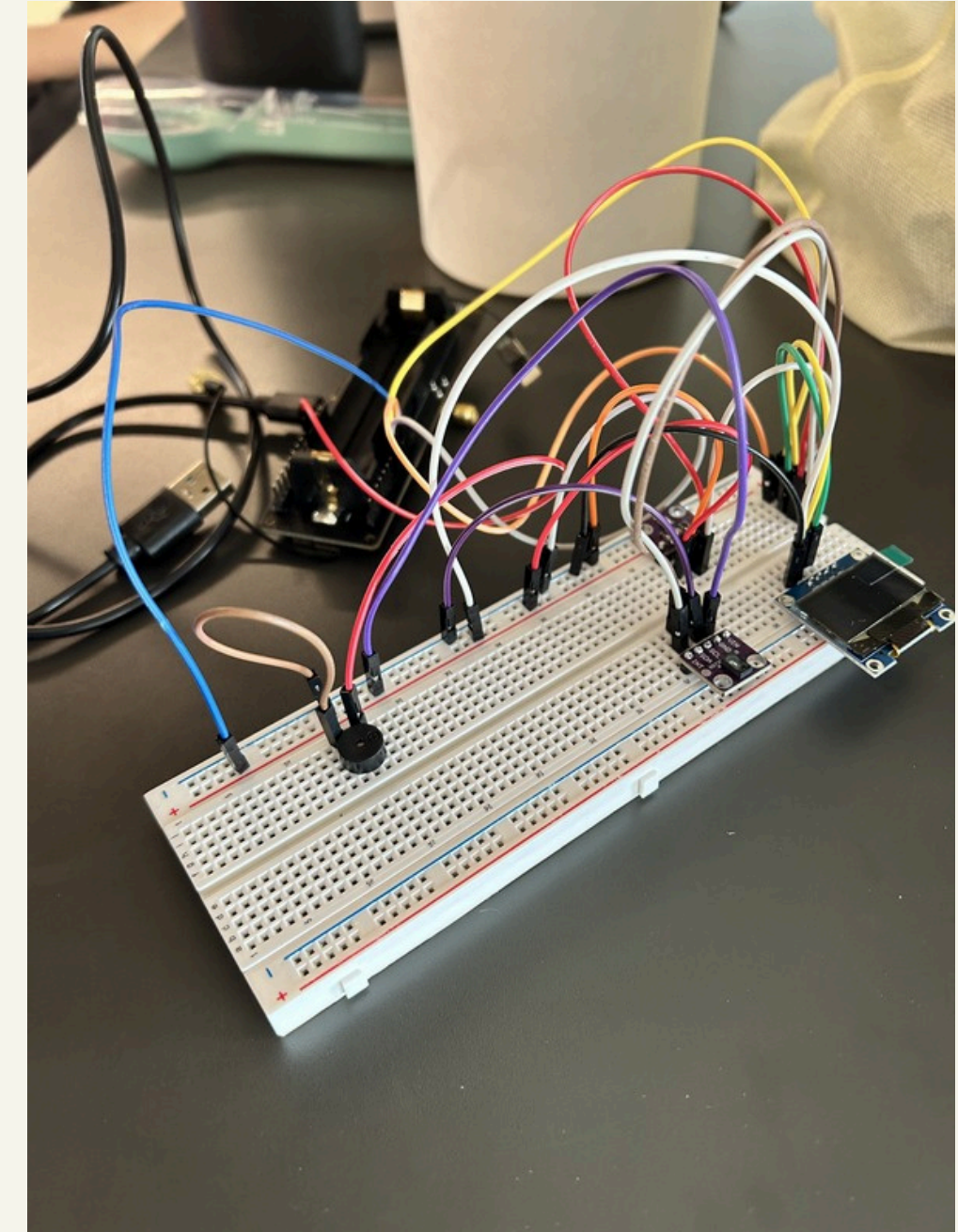
TESTING PROCESS

STEP 1: Testing Hardware

- Tested BME sensors (temperature, humidity, altitude, air pressure)
- Tested GPS location, LoRa com range

STEP 2: Testing 2 way communication

- Attempt to transfer the sensor data between CanSat and ground station





STEP 3: Full wiring + Parachute

- Began wiring everything on a breadboard
- Attach weights to the parachute to test fall speed + stability

STEP 4: Soldering + 3 way communication

- After testing breadboard we soldered the wires together and began testing communication between SOS Radio → Cansat → Ground Station





Final Assembly

- After gathering all the parts we put them together and tested the Cansat!
- This includes finished frame, solder, parachute, and assembling it.





THANK YOU!

Questions?

